

RUBIN SCOUT

Product Requirements Document

Multi-Messenger Astronomical Alert Platform

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Version: 0.1 | Date: May 2026
github.com/Namrata-Modha/rubin-scout

1. Executive Summary

Rubin Scout is an open-source multi-messenger astronomical alert platform built to aggregate, enrich, and surface transient events from global survey telescopes and gravitational wave detectors. It ingests alerts from the IAU Transient Name Server (TNS) and the ALeRCE broker, cross-references detections against SIMBAD, displays real telescope imagery from the Legacy Survey, and cross-matches optical transient candidates against LIGO/Virgo/KAGRA gravitational wave events from the GWTC catalog.

This document defines the product requirements for Rubin Scout v0.1 through v0.2, captures the stakeholder landscape, competitive positioning, gap analysis, and functional requirements for three planned feature additions: live GW event data from GWOSC, observatory visibility planning, and CHIME/FRB integration. It also defines the outreach strategy targeting ARIES (India) and Canadian institutions including the Dunlap Institute and McGill Space Institute.

2. Document Information

Document Title	Rubin Scout - Product Requirements Document
Version	1.0
Date	May 2026
Author	Namrata Modha, Full-Stack Developer
Status	Active Development - v0.1 Live
Live URL	rubin-scout.vercel.app
GitHub	github.com/Namrata-Modha/rubin-scout
Target Institutions	ARIES (India), Dunlap Institute (Toronto), McGill Space Institute (Montreal)

3. Product Overview

3.1 What Rubin Scout Does

Rubin Scout operates as a two-layer system: an ingestion backend that pulls and normalizes alerts from multiple astronomical data sources, and a React frontend that presents those alerts in an accessible, researcher-friendly interface. The platform is designed to reduce the manual effort required to contextualize a new transient detection against existing catalogs and gravitational wave events.

3.2 Current Data Sources

- IAU Transient Name Server (TNS) - primary transient discovery feed, daily CSV ingestion with photometry backfill via API
- ALeRCE Broker - ML classifications and g/r band light curves from ZTF via the official ALeRCE Python client
- SIMBAD - known object cross-matching via astroquery at 5 arcsecond cone radius
- Legacy Survey - real telescope cutout images constructed from RA/Dec coordinates
- LIGO/Virgo/KAGRA GWTC - gravitational wave event catalog (currently seeded, migrating to live GWOSC API)

3.3 Current Feature Set

- Alert dashboard with 994+ events, pagination, time filters (24h to All), confidence slider
- Classification filters: Supernova Ia, Supernova II, Nucleus, Black Hole Event, Kilonova, Nova, Blazar
- Alert detail page: telescope image, SIMBAD cross-match, g/r band light curve with error bars, observation count, confidence score
- All-sky map showing event distribution for current page
- GW Events page: GWTC catalog events with dominant classification, distance, sky area, optical counterpart search
- Optical counterpart search: PostGIS spatial query against ZTF alerts within GW 90% credible region
- Time-window fallback for poorly localized GW events
- Cone search REST API endpoint: GET /api/alerts/conesearch/query
- Full API documentation accessible at /api/docs

4. Stakeholder Analysis

Stakeholder	Role	Institution	Interest / Need
Dr. Kuntal Misra	ILMT Project Investigator	ARIES, India	GW counterpart follow-up, supernova alerts
Dr. Brajesh Kumar	ILMT Project Scientist	ARIES, India	Pipeline integration, operational workflow
Dunlap Institute	Research group	University of Toronto	Time-domain transient analysis, multi-messenger

McGill Space Institute	Research group	McGill University	FRB research, CHIME data analysis
CHIME/FRB Collaboration	Observatory team	DRAO, Penticton BC	FRB alert distribution, public catalog
Namrata Modha	Developer / Owner	London, ON, Canada	Portfolio, computational astrophysics career path

4.1 Primary Target: ARIES / ILMT Team

The International Liquid Mirror Telescope (ILMT) at Devasthal, Nainital is operated by ARIES and is India's first optical survey telescope performing nightly transient detection. ILMT's PyLMT pipeline detects transients and passes follow-up targets to the 3.6m Devasthal Optical Telescope for spectroscopy. The science team has publicly expressed interest in Big Data and AI/ML approaches for transient classification.

The workflow gap Rubin Scout addresses: when ILMT detects a new optical transient, determining whether a coincident gravitational wave event exists in the LIGO catalog currently requires manual lookup. Rubin Scout automates this cross-match and returns candidates with angular separation, classification, and confidence scores.

4.2 Canadian Institutions

CHIME, operated at DRAO Penticton BC, is one of the most prolific Fast Radio Burst detectors globally. Its public alert catalog represents a natural third messenger stream for Rubin Scout alongside optical transients and gravitational waves. Canadian university research groups at Dunlap (Toronto) and McGill work directly with CHIME data and multi-messenger astronomy, making them relevant targets for institutional outreach once FRB integration is complete.

5. Competitive Analysis

Feature	Rubin Scout	Astro-COLIBRI	Chirp / GW Apps
Optical Transients (ZTF)	Yes - via ALeRCE + TNS	Aggregates links only	No
Inline Light Curves	Yes - g/r band with error bars	Links to AAVSO externally	No
SIMBAD Cross-matching	Yes - inline on detail page	No	No
Legacy Survey Images	Yes - inline telescope cutout	Links to ALADIN externally	No
GW Events (LIGO/Virgo)	Yes - GWTC catalog	Yes - real-time alerts	Yes - primary focus
GW Optical Counterpart Search	Yes - PostGIS spatial query	No	No
Fast Radio Bursts (CHIME)	Planned - CHIMEFRB integration	Yes	No

Observatory Visibility Planning	Planned - Devasthal + DRAO	Yes	No
Public REST API	Yes - cone search + alerts	No	No
Open Source	Yes	No	Partial
Mobile App	No	Yes (iOS + Android)	Yes

5.1 Differentiation Summary

Astro-COLIBRI is the primary comparable platform. The core distinction is depth versus breadth. Astro-COLIBRI is a notification and aggregation hub that redirects users to external tools for data. Rubin Scout surfaces the data inline: light curves, SIMBAD associations, telescope images, and GW cross-match results are all visible without leaving the platform.

The second differentiator is the REST API. Astro-COLIBRI has no public programmatic interface. Rubin Scout exposes a cone search endpoint that external pipelines such as PyLMT can query directly, enabling workflow integration rather than just human-facing discovery.

Rubin Scout does not compete with Astro-COLIBRI on breadth or real-time latency. Those remain gaps. The positioning is specialist optical transient analysis with multi-messenger context, not general-purpose alert aggregation.

6. Gap Analysis

Gap	Current State	Required State	Priority
GW Events	Manual seed (6 events)	Live GWOSC API (90+ events)	High
Visibility	External tools required	Inline altitude chart	Medium
FRB Coverage	None	CHIME integration	Medium
Database Uptime	Cold starts on Supabase	Keep-alive mechanism	High
Sky Map Label	Shows current page, labeled All-Sky	Page scope clearly labelled	Low
Light Curve Axis	Raw MJD values	Human-readable UTC dates	Medium

7. Functional Requirements

7.1 Feature: Live GW Events from GWOSC

Business Requirement

GW event data must reflect the full published GWTC catalog rather than a manually maintained hardcoded list. New catalog releases must be picked up automatically without code changes.

Functional Requirements

- FR-GW-01: System shall fetch GW events from the GWOSC public API at <https://gwosc.org/eventapi/json/allevnts/> using httpx. No authentication required.
- FR-GW-02: GPS timestamps from GWOSC shall be converted to UTC datetime using `astropy.time.Time(gps, format='gps')`.
- FR-GW-03: A weekly scheduled job shall re-run `seed_gw_events()` to pick up new GWTC releases.
- FR-GW-04: If GWOSC is unreachable, system shall log the error and return an empty list without crashing startup.
- FR-GW-05: Prose descriptions shall be retained for GW170817, GW190521, GW200105, GW200115, and GW231123. All other events shall have `description=None`.

Constraints

- Existing `GWEvent` model and `cross_match_event()` logic shall not be modified.
- `GRACEDB_PUBLIC_URL` constant shall be retained.

Acceptance Criteria

- AC-GW-01: GWOSC API returns 90 or more events and all are stored in the `GWEvent` table.
- AC-GW-02: GPS timestamps convert correctly to UTC datetime with no off-by-one errors on known events (GW170817 = 2017-08-17).
- AC-GW-03: Weekly scheduler job runs without manual intervention and logs a success entry.
- AC-GW-04: If GWOSC is unreachable at startup, application starts normally with a logged warning.
- AC-GW-05: Prose descriptions are present for GW170817, GW190521, GW200105, GW200115, and GW231123. All other events have null description.

7.2 Feature: Observatory Visibility Planning

Business Requirement

Researchers using ILMT (Devasthal) or DRAO (Penticton) need to know whether an alert target is observable from their telescope tonight before committing to follow-up. This information is currently unavailable in Rubin Scout and requires external tools.

Functional Requirements

- FR-VIS-01: Backend shall expose `GET /api/alerts/{oid}/visibility` accepting lat, lon, elevation, and optional date parameters.
- FR-VIS-02: Endpoint shall compute hourly altitude, dark time window, moon separation, and observable flag using `astropy.coordinates` only.
- FR-VIS-03: Observable flag shall be true if target altitude exceeds 30 degrees for at least 1 continuous hour during astronomical dark time.
- FR-VIS-04: Observatory presets shall include Devasthal (default), DRAO Penticton, DAO Victoria, Mauna Kea, Paranal, La Palma, and Palomar. All coordinates must be verified before implementation.
- FR-VIS-05: Frontend shall display an altitude vs time chart with green above 30 degrees, red below, and shaded dark time region.
- FR-VIS-06: Default observatory selection shall be Devasthal.
- FR-VIS-07: Visibility data shall be fetched on user interaction, not on page load.

Constraints

- No new astronomy libraries. astropy is already installed.
- Observatory presets stored in backend/app/utls/observatories.py, not in a database table.

Acceptance Criteria

- AC-VIS-01: Altitude curve returns 24 hourly data points for a given target and observatory.
- AC-VIS-02: Observable flag is true for a known circumpolar target at Mauna Kea and false for a target below the horizon.
- AC-VIS-03: Default observatory in the dropdown is Devasthal with correct coordinates loaded.
- AC-VIS-04: Chart renders green above 30 degrees altitude and red below, with dark time shaded.
- AC-VIS-05: Visibility card does not trigger an API call until the user opens it.

7.3 Feature: CHIME/FRB Integration

Business Requirement

Fast Radio Bursts detected by CHIME at DRAO Penticton represent a third messenger stream relevant to multi-messenger follow-up. Adding FRB coverage strengthens the Canadian institutional outreach angle and differentiates Rubin Scout from tools that treat FRBs as secondary.

Functional Requirements

- FR-FRB-01: System shall ingest FRB detections from the CHIMEFRB public catalog. Exact API endpoint to be verified at implementation time.
- FR-FRB-02: FRB objects shall be stored with classification=FRB, broker_source=chimefrb, and dispersion measure in sub_classification as DM={value}.
- FR-FRB-03: CHIMEFRB ingestion shall run on the same scheduler interval as TNS and ALeRCE with its own IngestionLog entry.
- FR-FRB-04: Dashboard shall display an FRB filter chip with a distinct color.
- FR-FRB-05: FRB detail page shall display dispersion measure with a plain English explanation.

Constraints

- Existing ingestion patterns (upsert, error handling, IngestionLog) must be followed exactly.
- If CHIMEFRB API is unreachable, log and return 0. Do not crash.

Acceptance Criteria

- AC-FRB-01: At least one real CHIMEFRB event appears in the dashboard with classification=FRB.
- AC-FRB-02: Dispersion measure value is stored and visible on the FRB detail page with plain English explanation.
- AC-FRB-03: FRB filter chip appears on dashboard and correctly filters to FRB-only results.
- AC-FRB-04: IngestionLog shows a completed entry for each CHIMEFRB ingestion run.

- AC-FRB-05: If CHIMEFRB API returns a 500 error, scheduler continues running and logs the failure without crashing.

7.4 Feature: Database Keep-Alive

Business Requirement

Supabase free tier pauses the database after inactivity, resulting in a 30-second cold start that will cause researchers who receive the outreach email to hit a blank page. This is a credibility risk that must be resolved before any institutional outreach.

Functional Requirements

- FR-KA-01: Backend shall expose GET /api/health/ping returning {status: ok, timestamp: <utc now>}. No auth, no rate limit, no DB query.
- FR-KA-02: Scheduler shall run a SELECT 1 keep-alive query every 4 minutes to maintain the SQLAlchemy connection pool.
- FR-KA-03: External cron-job.org job shall ping /api/health/ping every 5 minutes. Setup instructions documented in KEEPALIVE.md at project root.

Acceptance Criteria

- AC-KA-01: GET /api/health/ping returns HTTP 200 with {status: ok} within 500ms.
- AC-KA-02: Application logs show a SELECT 1 keep-alive query firing every 4 minutes.
- AC-KA-03: After cron-job.org is configured, rubin-scout.vercel.app loads without a cold-start delay when accessed by a new user.
- AC-KA-04: KEEPALIVE.md exists at project root with step-by-step cron-job.org setup instructions.

8. Estimated Effort and Timeline

The following estimates assume a single developer working on Rubin Scout part-time. FR-KA should be completed first as it has the lowest effort and the highest credibility impact for any outreach activity.

Feature	FR Group	Estimated Hours	Tasks
Live GW Events	FR-GW	8 hours	GWOSC API integration, GPS-to-UTC conversion, weekly scheduler job, error handling
Visibility Planning	FR-VIS	16 hours	Altitude calculation (astropy), observatory presets, backend endpoint, frontend chart, dark time shading
CHIME/FRB Integration	FR-FRB	12 hours	API research and verification, ingestion service, upsert pattern, FRB filter chip, DM display on detail page
Database Keep-Alive	FR-KA	4 hours	Ping endpoint, SELECT 1 scheduler job, cron-job.org setup, KEEPALIVE.md
Total Estimated Effort		40 hours (~1 week sprint)	Single developer sprint. FR-KA should be completed first (lowest effort, highest credibility impact).

Recommended sprint order: FR-KA (day 1) > FR-GW (days 1-2) > FR-FRB (days 2-4) > FR-VIS (days 4-7). This sequencing ensures the tool is stable and data-rich before the most complex frontend work begins.

9. Non-Functional Requirements

- NFR-01 Security: All string inputs are length-limited. Classification filters validated against allowlist. OID validated against ZTF naming pattern. Parameterized queries only.
- NFR-02 Rate Limiting: Read endpoints limited to 60 req/min. Spatial queries limited to 30 req/min via slowapi.
- NFR-03 Accuracy: All claims made in institutional outreach email must be verifiable from the codebase. No feature shall be described as live unless the implementation confirms it.
- NFR-04 Performance: Visibility computation shall return within 3 seconds for a single target. GWOSC fetch shall complete within 30 seconds or timeout gracefully.
- NFR-05 Availability: Database keep-alive mechanism shall be in place before any institutional email is sent.

10. Institutional Outreach Strategy

9.1 ARIES / ILMT - India

Primary Contact	Dr. Kuntal Misra - kuntal@aries.res.in (ILMT Project Investigator)
CC	Dr. Brajesh Kumar - brajesh@aries.res.in (ILMT Project Scientist)
Hook	GW optical counterpart search + visibility planning defaulted to Devasthal
Prerequisite	GWOSC integration live, keep-alive in place, visibility feature deployed
Timing	After Features 7.1, 7.2, and 7.4 are complete

9.2 Canadian Institutions

Target 1	Dunlap Institute for Astronomy and Astrophysics - University of Toronto
Target 2	McGill Space Institute - McGill University
Target 3	CHIME/FRB Collaboration - DRAO Penticton, BC
Hook	CHIMEFRB integration + DRAO visibility preset + open-source API
Prerequisite	Feature 7.3 (CHIMEFRB) complete and data live in dashboard
Timing	After Feature 7.3 is complete

11. Success Metrics

- At least one reply from ARIES or Canadian institution acknowledging the tool
- GWOSC integration returning full GWTC catalog (90+ events vs current 6)

- CHIMEFRB alerts appearing in dashboard with correct classification and DM values
- Visibility chart rendering correctly for Devasthal and DRAO Penticton
- Zero cold-start incidents after keep-alive is deployed
- GitHub repository public and code accurately reflects all feature claims in outreach email

12. Assumptions and Constraints

- Supabase free tier will remain in use. No budget for paid tier. Keep-alive via external cron is the mitigation.
- GWOSC public API requires no authentication and will remain publicly accessible.
- CHIMEFRB public catalog API endpoint must be verified at implementation time. Exact URL not confirmed at time of writing.
- astropy is the only astronomy library available. healpy is not installed (Windows-incompatible per existing code comments).
- Live GraceDB access (for O4 real-time events) requires LIGO member credentials and is out of scope for v0.1.
- The tool is not affiliated with Vera Rubin Observatory. Name is a homage to Vera Rubin. Disclaimer already present in footer.

Appendix: Technology Stack

Backend	FastAPI, Python, SQLAlchemy (async), PostgreSQL + PostGIS
Frontend	React, deployed on Vercel
Database	Supabase (PostgreSQL, free tier)
Data Sources	TNS, ALeRCE, SIMBAD (astroquery), Legacy Survey, GWOSC
Planned Sources	CHIMEFRB public catalog
Astronomy Libraries	astropy, astroquery
Scheduling	APScheduler (AsyncIOScheduler)
Security	slowapi rate limiting, input validation, parameterized queries
Version	v0.1 - live at rubin-scout.vercel.app